Extension: Contextual Thompson Sampling

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Extends the MVP bandits (§5 of `mvp_specification.tex`) by conditioning action selection on the current activity level.

1 Formulation

Context $c_t = s_t.\texttt{activity} \in \{\texttt{sedentary}, \texttt{active}\}$. Instead of one (α_a, β_a) per action, maintain $(\alpha_{c,a}, \beta_{c,a})$ for each context-action pair. Sample $\theta_{c_t,a} \sim \text{Beta}(\alpha_{c_t,a}, \beta_{c_t,a})$, pick $a^* = \arg\max_a \theta_{c_t,a}$. Update only the observed (c_t, a_t) posterior.

2 Optimal Policy

The optimal contextual policy (from the transition matrices in §3) is: nudge when sedentary ($0.3 > 0.1$), idle when active ($0.6 > 0.5$). Stationary distribution: $\pi_s = 4/7$, $\pi_a = 3/7$.

$$E[r] = \frac{4}{7} \cdot 0.3 + \frac{3}{7} \cdot 0.6 = \frac{3}{7} \approx 0.429 \text{ per step} \implies 192.9 \text{ total}$$

Best non-contextual (always nudge): $E[r] = 0.375$, total 168.8. The contextual bound is 14.3% higher.

3 Results

50 seeds, 450 timesteps. Full methodology follows §6 of `mvp_specification.tex`.

Agent	Total Reward	Per Step	vs Contextual Opt
Contextual TS	176.7 ± 15.2	0.393	-8.4%
Standard TS	158.5 ± 12.6	0.352	-17.8%
Contextual optimal	192.9	0.429	—
Non-contextual optimal	168.8	0.375	—

Per-step reward in the last 50 steps: contextual TS reaches 0.422 (1.5% from optimal), standard TS at 0.369. The agent converges within the episode.

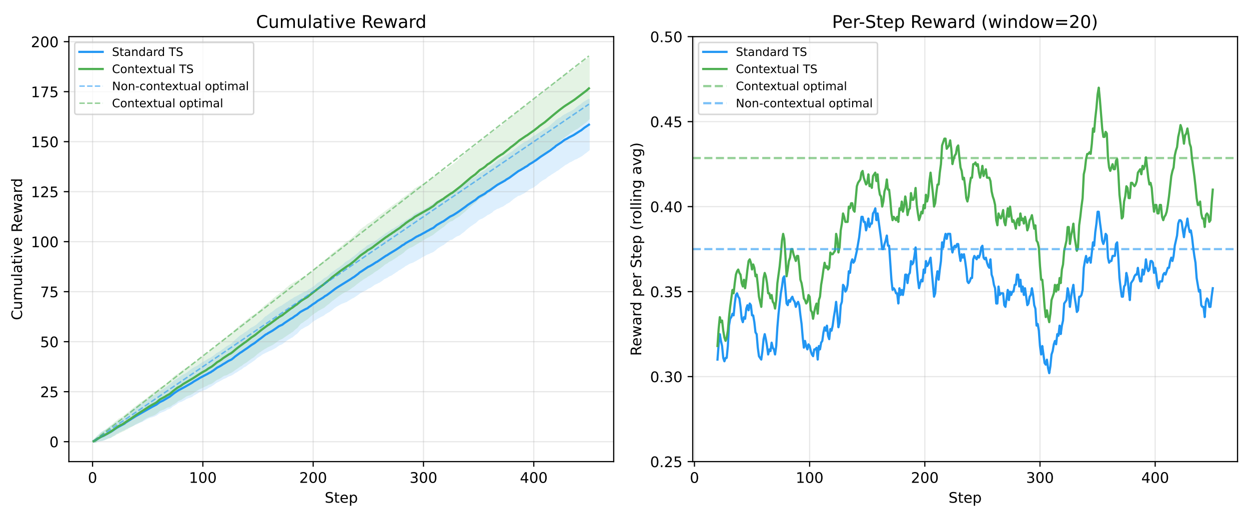

Figure 1: Left: cumulative reward (50 seeds, ± 1 SD). Right: per-step reward (20-step rolling avg). Contextual TS converges toward $3/7 \approx 0.429$; standard TS plateaus at 0.375.